Prediction for August

Desmos Mastery (WH)

SAT Math Practice Set

1.

$$f(x) = x^2 - 42x + 80$$

The function f is defined by the given equation. Which of the following equivalent forms of the equation displays the zeros of the function as constants or coefficients?

A) $f(x) = x^2 - 5x - 37x + 80$

B) $f(x) = (x - 2)(x - 40)$

C) $f(x) = x(x - 42) + 80$

D) $f(x) = (x - 21)^2 - 361$

2.

$$f(x) = k(1.83)^x$$

The function f is defined by the given equation, where k is a constant. The value of $f(x)$ increases by $p\%$ for every increase of x by 1. For which of the following functions, where k is a constant, does the value of $g(x)$ increase by $p\%$ for every increase of x by $\frac{1}{4}$?

A) $g(x) = k\,(1.83^x)^{1/4}$

B) $g(x) = k\,(1.83^x)^4$

C) $g(x) = k(1.83)^{x+1/4}$

D) $g(x) = k(1.83)^{x-1/4}$

3. A line intersects two parallel lines, forming four acute angles and four obtuse angles. The measure of one of the acute angles is $(9x - 490)^\circ$. The sum of the measures of one of the acute angles and three of the obtuse angles is $(-18x + w)^\circ$. What is the value of w?

Answer:_____

4. The exponential function f is defined by $f(x) = ab^x$, where a and b are positive constants. If $f(n-1) = f(n) + \frac{82}{100}f(n-1)$, where n is a constant, what is the value of b?

Answer: _____

5.

$$f(x) = 3(x - a)(x - b)(x - c)$$

The function f is defined by the given equation, where a, b, and c are distinct constants. When $a < x < b$, the value of $f(x)$ is positive. The graph of $y = f(x)$ in the xy-plane contains the point (r,s), where r and s are constants. If $s = 8$, which of the following could be true?

I. $r < a$

II. $b < r < c$

III. $r > c$

A) I only

B) III only

C) I and III only

D) II and III only

6. The perimeter of an equilateral triangle is 642 centimeters. The three vertices of the triangle lie on a circle. The radius of the circle is $w\sqrt{3}$ centimeters. What is the value of w?

Answer: _____

7. In triangle *FGH* and triangle *KLM*, the measures of angles *G* and *L* are each 60°. The lengths of *GH* and *LM* are each 34 centimeters, and $\frac{GH}{GF} = \frac{LM}{LK}$. Which additional piece of information would be necessary to prove that triangle *FGH* is congruent to triangle *KLM*?

A) The lengths of *FG* and *KM* are each 17 centimeters.

B) The measures of angles *H* and *M* are each 30°.

C) The measures of angles *F* and *K* are each 90°.

D) No additional information is necessary.

8. A square map has a side length of 35 inches, and 1 inch on the map represents an actual distance of 17 miles. A smaller version of the same map is printed as a square with the side length 55% shorter than the side length of the previous map. On the smaller map, which of the following is closest to the actual distance, in miles, represented by 1 inch?

A) 7.65

B) 10.97

C) 19.25

D) 37.78

9.

$$2\cos(90 - a)^\circ \cdot \cos b^\circ + \sin(a + y)^\circ \cdot \sin(90 - b)^\circ$$

In the given expression, a and b are constants such that $\sin a^\circ = 0.50$, $\cos b^\circ = 0.99$, and $0 < b < 90$. What is the value of the given expression when $y = 0$?

Answer: _____

10. Two identical rectangular prisms each have a height of 90 centimeters (cm). The base of each prism is a square, and the surface area of prism is K cm^2. If the prisms are glued together along a square base, the resulting prism has a surface area of $\frac{92}{47}K \text{ cm}^2$.

What is the side length, in cm, of each square base?

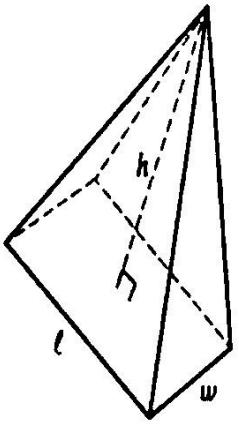
A) 4

B) 8

C) 9

D) 16

11.



Note: Figure not drawn to scale.

The figure shown is a right rectangular pyramid, where ℓ = 18 units, w = 9 units, and h = 12 units. What is the surface area, in square units, of the pyramid?

Answer: _____

12. During the first 8.00 seconds after a car started moving, its speed increased to 11.0 meters per second. From 8.00 seconds to 14.0 seconds after the car started moving, its speed increased from 11.0 meters per second to 23.0 meters per second. To the nearest hundredth, what is the positive difference between the average rate of change of the speed of the car, in meters per second per second, during the first 8.00 seconds after the car started moving and the average rate of change of the speed of the car, in meters per second per second, from 8.00 seconds to 14.0 seconds after the car started moving?

Answer: _____

13.

$$x^2 + y^2 - 6x - 10y - 4n = 0$$

The given equation represents circle A in the xy-plane, where n is a constant. Point (5,7) lies on circle B, which has the same center but twice the diameter as circle A. What is the value of n?

Answer: _____

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14. A car dealership has only sedans, SUVs, and minivans for sale. On Monday, 30% of the vehicles for sale were sedans and 50% were SUVs. If there were 57 sedans for sale at the dealership on Monday, how many minivans were for sale?

Answer: _____

15. For a party, 65 hamburger buns are needed. Hamburger buns are sold in packages of 12. What is the minimum number of packages that should be bought for the party?

Answer: _____

16. In triangle *XY Z*, the measure of angle *X* is x°, the measure of angle *Y* is y°, and the measure of angle *Z* is 31°. If $x > y > 31$, which of the following additional pieces of information is NOT sufficient to determine the values of *x* and *y*? *[*Bonus Question]*

A) The value of $31 - x - 3y$

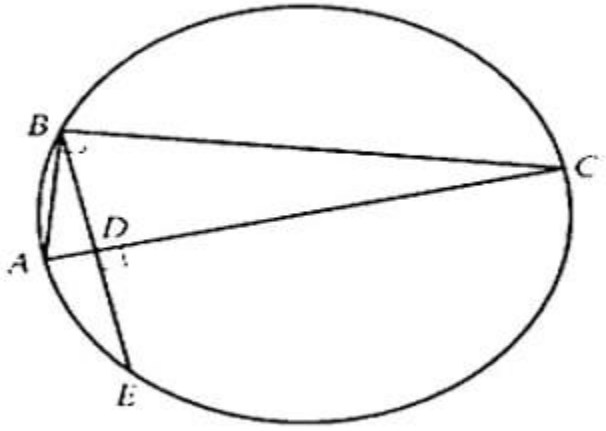
B) The value of $31 + 3x + y$

C) The value of $31 - 3x + 3y$

D) The value of $31 - 3x - 3y$

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17.

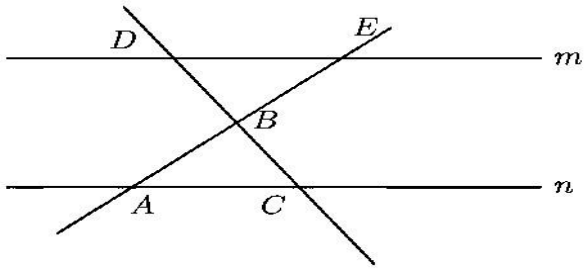


Note: Figure not drawn to scale.

In the figure shown, points *A*, *B*, *C*, and E lie on the circle, and $AB < BC$. Segment *AC* is perpendicular to segment *BE* at point *D*, and $BD = \sqrt{346}$. The diameter of the is 175. If $\frac{CD}{AD} = r$, what is the value of r?

Answer: _____

18.



Note: Figure not drawn to scale.

In the figure, line *m* is parallel to line *n*, and lines *AE* and *CD* intersect at point *B*. Which additional piece of information is sufficient to prove that triangle *ABC* is congruent to triangle *EBD*?

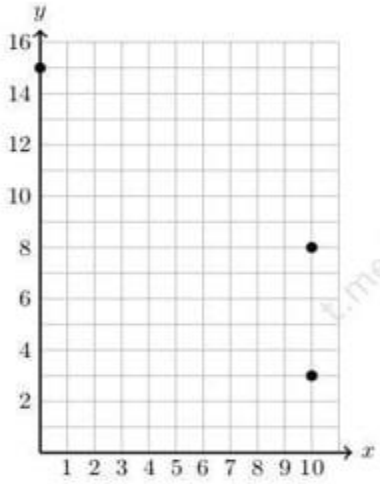
A) AB = 16 and DB = 16.

B) AB = 16 and EB = 16.

C) Triangles ABC and EBD are isosceles.

D) No additional information is necessary to determine that the two triangles are congruent.

19. The points (0,15), (10,8), and (10,3) are shown in the *xy*-plane, where the *x*-axis and *y*-axis are measured in units. These points define one of the bases of a triangular prism. The distance between the two bases of the prism is 20 units. What is the volume, in cubic units, of the prism?



Answer: _____